

Earth-based unmanned solar sintering system for lunar bricks with regolith simulant

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ABSTRACT

The concentrated sintering in-situ construction method, using solar energy as the energy source and lunar regolith as the raw material, offers a promising approach for lunar brick production. This paper introduces an unmanned solar-powered sintering system to fabricate lunar bricks using a regolith simulant. The performance of the concentrating module was analyzed through numerical simulations. An integrated experimental prototype was then deployed, and experiments were conducted form simulated lunar regolith bricks with adequate strength. The overall performance and optimization potential of the system were evaluated. The automated experimental platform and machinery, designed to operate under lunar conditions, provide valuable insights for the advancement of extraterrestrial construction technologies and equipment.

1. Introduction

Lunar exploration missions have undergone extensive efforts over time, yielding significant results. The construction of bases on the lunar surface has emerged as a strategic priority for many nations [1,2]. In this context, lunar regolith is an abundantly available material, and solar energy is a uniquely convenient energy source. Utilizing solar energy for the direct sintering and formation of lunar regolith presents a highly feasible in-situ construction strategy with substantial potential applications [1,3].

However, most existing experimental machines designed for solar forming of simulated lunar soil feature simplistic structures and lack advanced sensing and motion systems. Many employ molten powder bed forming, which is challenging due to the weak bonding forces between simulated lunar regolith particles. Developing automated experimental machinery that employs molding and pressurization techniques can enhance the bonding conditions of lunar regolith particles during the sintering process, resulting in components with superior performance.

Although solar energy is readily accessible, it exhibits temporal variability and spatial inhomogeneity, particularly in the domain of

concentrated light. Numerous researchers have conducted studies aimed at optimizing the optical performance and structural design of solar energy concentration systems. Nonetheless, there remains a gap in the specific area of simulating lunar regolith sintering. Thus, a focused analysis of solar concentrating systems in the context of lunar surface construction is warranted to explore the underlying mechanisms and potential optimization pathways.

Fresnel lenses, characterized by their complex optical structures, present considerable opportunities for research. Although many scholars have explored the parameter optimization of Fresnel lenses, the majority of applications have concentrated on direct imaging or thermal concentration. Research specifically focused on leveraging the light-collecting capabilities of Fresnel lenses for lunar surface construction is limited. Given that the melting point of simulated lunar regolith typically exceeds 1000 degrees Celsius, the heat collection performance of Fresnel lenses must be rigorously evaluated. Therefore, a systematic design of a device to facilitate the sintering of simulated lunar regolith using Fresnel lenses is essential, as well as a thorough analysis and optimization of their heat collection performance in this scenario.

In this study, a simulated lunar regolith solar sintering

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device—referred to as the unmanned solar sintering system (USSS)—was designed and developed for potential application in lunar surface construction, as illustrated in Fig. 1. The system is capable of autonomously tracking the sun's movement and employs Fresnel lenses to concentrate solar energy, generating sufficient heat to sinter the simulated lunar regolith within a mold. Furthermore, the performance and optimization of the USSS were evaluated through simulation and experimental analyses, providing guidance for future improvements of the device.

This work is organized as follows: Section 2 reviews the current status of research. Section 3 describes the system design of USSS. Section 4 models the system analysis based on simulation. Section 5 discusses the impact of different design parameters on system performance. Section 6 verifies the functionality of USSS based on deployment integration and experimental testing. Section 7 summarizes the research and suggests future work.

2. Related work

This section reviews the state-of-the-art research in three key areas: (1) experimental systems for solar sintering of lunar regolith simulants, (2) solar concentration technologies, and (3) design optimization of Fresnel lenses. It highlights gaps in automated systems and lunar-specific applications, establishing the foundation for the proposed USSS design.

2.1. Experimental machine for solar sintering of regolith simulant

Various organizations and researchers worldwide have attempted to construct experimental machines for solar sintering of simulated lunar regolith, which can be categorized based on their application scenarios. One such scenario is dust mitigation; lunar dust not only disrupts surface operations but also poses health risks to astronauts [4–6]. Dust curing employs concentrated sunlight to melt and reform a layer of lunar regolith. For example, NASA's Goddard Space Flight Center has developed a dust mitigation vehicle (DMV) capable of transforming loose lunar regolith into a hardened, dust-free platform [7]. This method of curing lunar soil also served as an important reference for the manufacturing method of lunar landing pads. [8–10]

Beyond dust curing, the application of concentrated sunlight to fabricate three-dimensional components resembling lunar regolith is a widely researched area. The simplest approach uses lenses to directly concentrate light for sintering. For example, Germany's OSRAM utilized

xenon lamps to simulate sunlight, conducting preliminary simulations of lunar regolith formation, followed by iterative upgrades [11]. The Liquifier Systems Group proposed a mobile solar concentrator that can achieve XYZ three-axis movement during the forming process [12]. Moreover, alternative methods, such as using optical fibers to channel sunlight to the forming area [13–16], have been explored. The United States has developed and deployed an optical waveguide (OW) system at the ISRU simulation test site on Mauna Kea Island in Hawaii [13]. Additionally, the use of mirrors to converge light without lenses, akin to a “solar cooker” employed on Earth [14], represents another viable approach, exemplified by NASA's Cassegrain concentrator [17].

In conclusion, while several experimental devices for simulating the sintering of lunar regolith using concentrated light have been developed, most remain at a basic construction level. There is a notable absence of comprehensive scientific investigations and integrated deployments of automatic light-tracking functions to adapt to solar motion and effective heat gathering modules. Furthermore, many efforts in simulated lunar regolith sintering rely on molten powder bed forming processes, which face challenges due to the weaker interlayer bonding of simulated lunar regolith, complicating component formation and reducing strength. In contrast, enhancing the compaction of lunar regolith through molded pressurization can improve interparticle bonding during sintering. Thus, designing a complete experimental system for deployment can substantially enhance the quality of formed components.

2.2. Concentration analysis of solar energy

Solar concentration technology focuses sunlight from a large area into a smaller area through reflection or refraction, widely applied in solar thermal power generation, photovoltaic systems, and solar water heaters. This technology has attracted considerable attention from researchers. The analysis of solar energy concentration can be approached from various perspectives. One aspect focuses on enhancing material properties, such as improving transmittance through the development of optical materials with high reflectivity or transmittance [18]. For instance, ren et al. used a new PMMA material to obtain higher transmittance [19], in addition to setting up a transmittance-enhancing material on top of the original optical material is also an important means [20,21]. Another direction is to optimize the system design. Mayank et al. Choosing the right design and geometry for the light harvesting system will reduce the dependence on electrical loads and thus improve its environmental friendliness [22]. In addition, integrated solar tracking allows real-time reorientation of the machine to improve

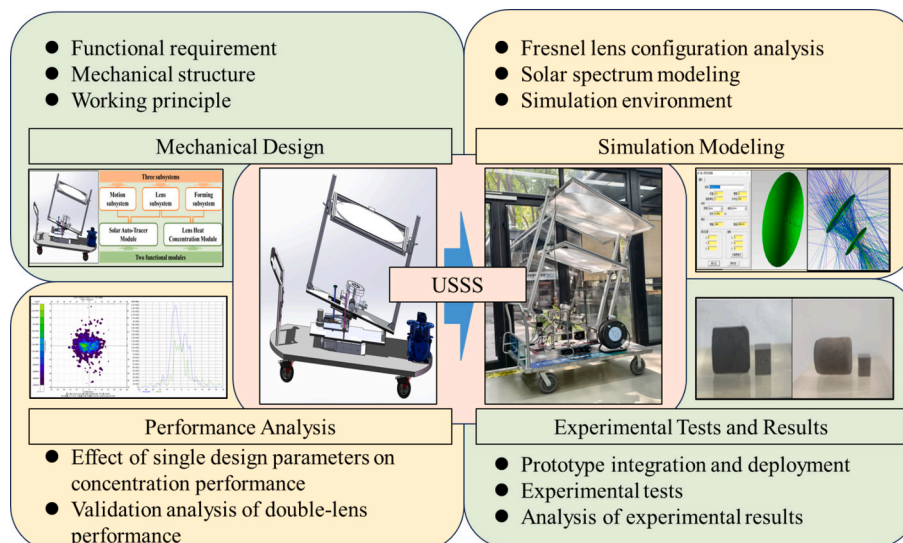


Fig. 1. Abstract graph.

energy input. In addition, optical configurations can be optimized [23–25]. Wang et al. analyzed the relationship between concentrator mounting angle and concentration ratio and energy efficiency based on numerical simulations [26]. Spectrally separated focusing techniques can also enhance the application of different wavelengths of light to improve the overall efficiency of the system [27,28].

In summary, while solar concentrator technology has been extensively utilized in photovoltaic power generation and thermal applications, its analysis in the context of solar shaping on the lunar surface remains relatively rudimentary. On one hand, the factors influencing and optimizing the concentration system warrant further exploration. On the other hand, integrating system functions such as solar tracking could significantly enhance the solar forming process for simulated lunar regolith.

2.3. Optimized design of Fresnel lenses

Due to its distinctive curved surface and textural composition, numerous researchers have examined the parameter optimization of Fresnel lenses for various applications. The specific application scenarios are analogous to those of conventional convex lenses, which can be broadly categorized into two primary areas: imaging and thermal collection. In the imaging domain, the most direct method for optimizing Fresnel lens parameters to enhance image quality is improving the lens's transmittance, thereby minimizing light loss within the lens medium. For example, Chang's research utilized the NSGA-II algorithm and Kriging model to optimize the process parameters of Fresnel lenses, enhancing their transmission performance [29]. Meanwhile, Delullier et al. concentrated on the refractive index of the lenses to elevate their efficacy in mid-infrared imaging applications [30]. Beyond improving transmittance, attention should also be directed toward the unique curved surfaces and textural properties of Fresnel lenses. For example, Karel et al. proposed a continuous free-form surface design algorithm tailored for Fresnel lens surfaces [31].

Regarding thermal conduction, the primary application is the concentration of solar energy for the thermal treatment of materials. Notably, Lina et al. proposed a hybrid system optimized through genetic algorithms, which enhances thermal conduction capabilities [32]. Garcia et al. developed a three-dimensional elliptical Fresnel lens, optimizing the number of grooves within the lens to achieve superior solar flux concentration [33]. Additionally, Ma et al. derived differential equations outlining the ideal Fresnel lens shape for optimization [34]. Beyond direct heating applications, Fresnel lenses are extensively employed in photovoltaic systems [35–38]. Minglei et al. created an algorithm for modeling and optimizing the parameters of planar focusing Fresnel lenses specifically designed for microprisms [39]. Similarly, leveraging their light-concentrating capabilities, Fresnel lenses can also be utilized in optical illumination [40–43]. Kumar et al. proposed a Fresnel lens design based on average wavelength, which enhances optical efficiency and uniformity while regulating the temperature at the focal plane [44]. In addition, some scholars have explored the overall optimization of multiple Fresnel lenses or the integration of a Fresnel lens with other optical components [45,46].

In summary, while numerous scholars have explored the parameter optimization of Fresnel lenses, most have concentrated on direct imaging or thermal collection applications. There is a notable scarcity of studies focusing on the utilization of Fresnel lenses' light-collecting capabilities for lunar surface construction. Therefore, a systematic analysis and optimization of Fresnel lenses' thermal gathering performance in this context is warranted.

3. Design of USSS

This section details the architecture of the Unmanned Solar Sintering System (USSS), including functional requirements for lunar operations, mechanical subsystems (motion, lens, forming), and operational

principles for automated solar tracking and sintering.

3.1. Functional requirement

In terms of motion capabilities, the system must possess sufficient attitude adjustment capacity to maintain the sun's rays perpendicular to the surface of the incident lens, as illustrated in Fig. 2. The movement of the sun's rays can generally be decomposed into changes in the altitude angle and the azimuth angle. The altitude angle represents the angle between the sun and the horizon during its motion, characterizing the vertical position of the sun's rays relative to the ground. This angle typically reaches its maximum value at noon, with a theoretical range of 0–90 degrees. The azimuth angle indicates the sun's horizontal direction relative to the ground receiver, generally measured from true north, with a theoretical range of 0–360 degrees. To accommodate variations in solar altitude and azimuth angles, the motion of the ground experimental system must also be decomposed.

As shown in Fig. 2, the change in the sun's altitude angle necessitates that the device's lens plane executes pitching motions to align with the geometric relationship. The pitch angle of the device is the cosine of the sun's altitude angle. Specifically, at noon, when the sun's altitude angle is approximately 90 degrees, the lens pitch angle approaches 0 degrees, indicating a nearly horizontal position. Under optimal sunlight conditions at noon, the horizontal component of the lens load is minimized, while the vertical component reaches its maximum, resulting in optimal load support for the device. Therefore, the system's conditions are also optimal under these circumstances, facilitating subsequent experimental developments. For changes in solar azimuth, the entire device rotates around its base to adapt.

The pitching motion entails significant adjustments to the system's attitude, warranting a specific examination of its travel requirements. For example, in Wuhan, China, the solar altitude angles for the four key solar terms—spring equinox, summer solstice, autumn equinox, and winter solstice—are presented in Table 1. This analysis considers only the angle changes during daytime hours from 10:00 to 16:00. Given that Wuhan's latitude exceeds that of the Tropic of Cancer, the solar altitude angle during the summer solstice varies between 42 and 81 degrees. According to the angular relationship between the solar altitude and pitch angles described previously, the pitch angle should range from 9 to 48 degrees. The vernal and autumnal equinoxes exhibit similar solar altitude angle changes, ranging from 32 to 59 degrees, indicating that the pitch angle should be between 31 and 58 degrees. During the winter solstice, the solar altitude angle is comparatively low, ranging from 15 to 36 degrees, which suggests that the pitch angle should be between 54 and 75 degrees. To optimize lighting conditions, the experimental operations of this ground system will primarily occur in summer; thus, considering seasonal variations and structural limitations, the minimum design pitch angle should be approximately 8 degrees, while the maximum should be around 45 degrees. It is essential to acknowledge

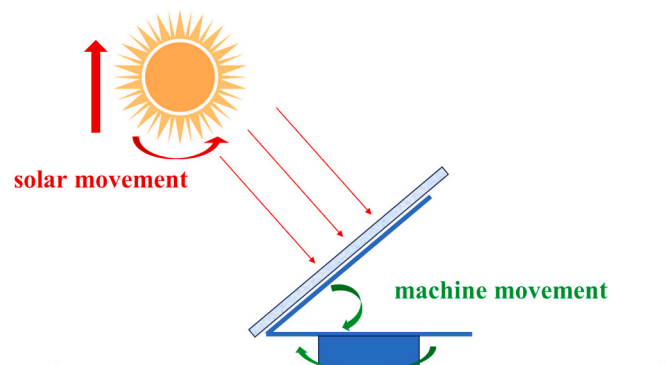


Fig. 2. Motion requirements for machine.

Table 1
Solar altitude angle on a typical day (In the case of the Wuhan area).

	March 21	June 22	September 23	December 22
10:00	43	57	46	26
11:00	53	70	55	33
12:00	59	81	59	36
13:00	59	79	58	35
14:00	53	68	51	31
15:00	43	55	41	24
16:00	32	42	29	15

that due to the limitations of the experimental conditions in this study and the intention to conduct most experimental work in Wuhan to mitigate equipment damage risks during transportation, the solar conditions in Wuhan are emphasized in the system design.

In terms of environmental factors, low gravity, vacuum, and dust are the main characteristics of the lunar environment. Among these, vacuum conditions significantly affect the temperature and composition changes during material melting. Therefore, vacuum conditions are a critical sintering condition that must be considered in device design. The effects of low gravity and dust are secondary and difficult to simulate effectively under ground conditions, so they were simplified in the design.

3.2. Mechanical structure

The overall design of the system is depicted in Fig. 3. The experimental system resembles a trolley, and this design is primarily intended to facilitate the movement of equipment before and after experiments. The core working mechanism is mounted on the upper portion of the trolley. The ground experimental system can be segmented into three subsystems based on its composition: the motion subsystem, the lens subsystem, and the forming subsystem. In functional terms, it can be categorized into two main modules: the solar energy automatic light-tracking module and the lens heat-gathering module.

3.2.1. Motion subsystem design

The motion subsystem primarily governs the upper component of the bracket plate, enabling dual-axis motion of pitch and rotation to satisfy the attitude requirements for ray tracing. The pitching motion of the subsystem relies on vertical elevation and motion conversion. As illustrated in Fig. 4, a motor on the right side controls the lift table's vertical movement via a belt drive wheel, with two vertical guide columns securing the lift table's degrees of freedom. This lifting motion is then converted to the bracket plate through a slider, pivoting at the pitch hinge. Furthermore, the motor is equipped with a gear reducer, which amplifies torque to ensure sufficient jacking force while preventing reverse rotation and avoiding potential falls of the lift table. The rotary motion is managed by a bottom rotary platform connected to the turntable plate, synchronizing the rotation of all upper mechanisms. The rotary platform employs stepping motors to facilitate both clockwise and

counterclockwise motion, providing the necessary driving force.

3.2.2. Lens subsystem design

The lens subsystem primarily depends on the light sensor to regulate the orientation of the lens, followed by the Fresnel lens that focuses the light to create a high-heat spot for the sintering process. As illustrated in Fig. 5, it comprises four main components: the light sensor mounting plate, the Fresnel lens, the height module, and the vertical support. The light-sensitive mounting plate maintains a parallel relationship with the lens to ensure precise adjustment of the light direction; the Fresnel lens is circular, tangent to the surrounding frame, and securely embedded; the height module is integrated into the vertical support and can be adjusted vertically to accommodate the focal length; it features a manual fixing bolt that secures the vertical support after the lens is focused, thereby maintaining the lens height.

3.2.3. Forming subsystem design

The forming subsystem is designed to hold the molds and connect various monitoring or auxiliary equipment to facilitate the sintering and shaping of components. As depicted in Fig. 6, its main body is a pure metal vacuum tank with an 80 mm vacuum glass opening at the top for spot projection, and an internal forming chamber for mold placement. In addition, the vacuum tank includes several interfaces for connecting external equipment. The vacuum gauge and pump interfaces share a branch channel that connects to the vacuum pump, ensuring a vacuum environment within the tank. The thermocouple interfaces, both internal and external, are equipped with contact points: the internal connection interfaces with the thermocouple meter for temperature measurement of the mold, while the external connection interfaces with a temperature display panel for monitoring. The atmosphere interface has two nozzles connected to a gas pump to create an inert gas environment during the sintering process.

3.3. Working principle

The working principle of the ground test system designed in this study is illustrated in Fig. 7. In terms of control and drive, the light sensor first captures sunlight, and based on the variable resistance of the photoresistor array, the light signal is converted into an electrical signal and transmitted to the core control board. Subsequently, the control board issues commands to the pitch motor and the rotary motor based on the signal data. During command execution, the light sensors continuously collect light signals, and the motors are driven by constant signal feedback, allowing for continuous attitude adjustments until the sunlight is vertically incident on the lens incidence surface. In the concentrated light forming process, the adjusted attitude of the lens converges sunlight to create a high heat spot, which then enters the mold surface in the forming cavity of the vacuum tank. The graphite mold's strong light absorption and thermal conductivity convert light energy into heat energy, simulating the sintering of lunar soil and curing within the mold. The entire sintering environment is maintained under vacuum

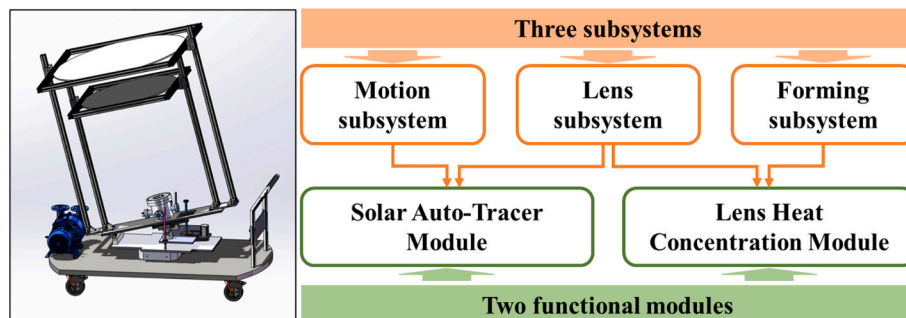


Fig. 3. Overall design.

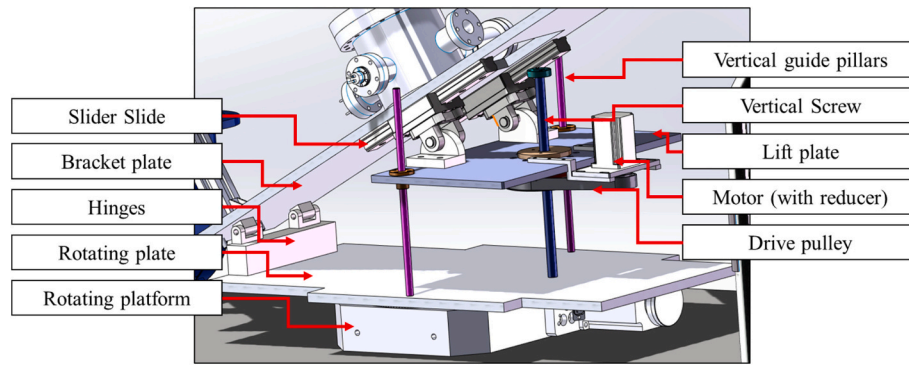


Fig. 4. Motion subsystem design model.

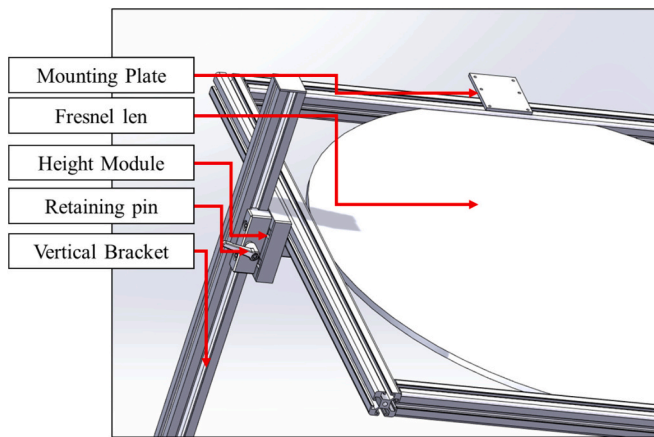


Fig. 5. Lens subsystem design model.

by a vacuum pump. Furthermore, this ground test system is equipped with monitoring devices, including thermocouple monitoring and vacuum monitoring. The thermocouple monitoring employs a contact method that directly inserts into the graphite mold to ensure accurate and real-time temperature measurements. The vacuum level is monitored throughout the sintering process to maintain stable environmental conditions.

4. Simulation modeling

Numerical models for optical concentration are established, covering Fresnel lens configuration, solar spectrum simplification (AM1.5 standard), and TracePro-based simulation environments for light-path analysis.

4.1. Fresnel lens configuration analysis

The Fresnel lens is an adaptation of the conventional convex lens developed by the French physicist and engineer Fresnel, based on the lens's characteristics [47,48]. As shown in Fig. 8, Fresnel aimed to reduce the excess thickness of the lens along the axial direction while preserving the curvature of the main focusing point, thereby constructing a new lens type with improved focusing performance. Visually, a Fresnel lens consists of a series of annular bands, with the configuration of each band determined by three elements: radius of curvature, coordinates of the center of curvature, and angle of interception.

The radius of curvature dictates the size and shape of each ring band, calculable using the following expression:

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad (1)$$

where f is the focal length of the lens element, n is the optical refractive index of the lens, and R_1 and R_2 are the curvatures of the two sides of the lens element, respectively.

For a general Fresnel lens, the incident surface is flat, so R_2 tends to infinity, which is obtained by simplification:

$$R = f(n - 1) \quad (2)$$

The coordinates of the center of curvature determine the position of each ring band. Given that the ring bands of the Fresnel lens are equidistant from one another (the equidistant distance is the ring spacing), the non-axial coordinates can be derived by simply assembling individual ring bands without additional calculations. Thus, attention is focused on deriving the axial coordinates of the curvature circle. And the Y-axis coordinate site(i) of the center of the i th annular band can be calculated from its radius of curvature $R(i)$ and lens diameter r :

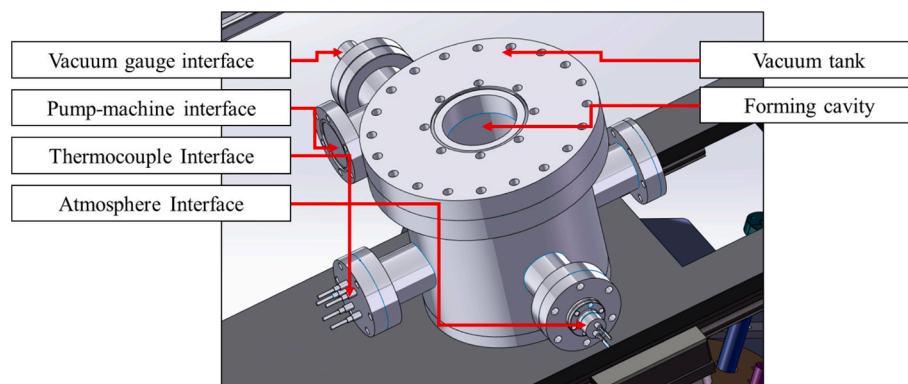


Fig. 6. Forming subsystem design model.

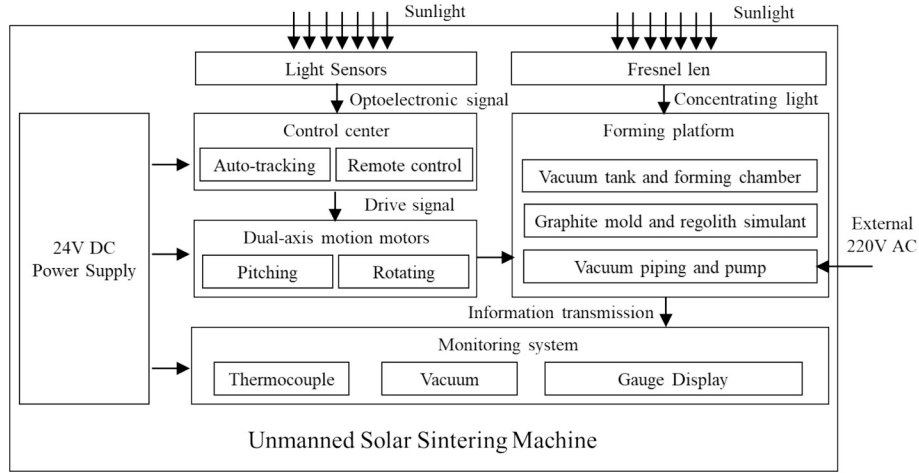


Fig. 7. Working principle block diagram.

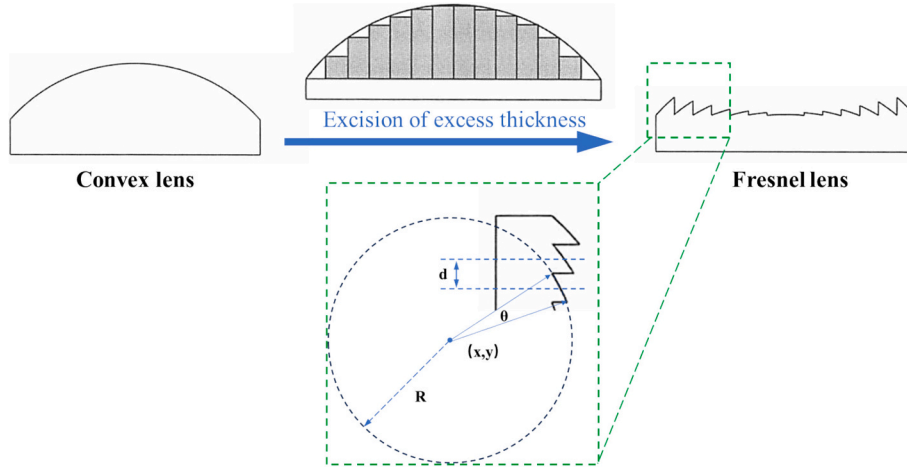


Fig. 8. Principle of construction of Fresnel lens.

$$site(i) = -\sqrt{R(i)^2 - r^2} \quad (3)$$

The intercept angle of the curvature circle is the value of the start angle θ_{1} and the end angle θ_{2} of each ring band, which can be derived from the ring spacing d , the lens radius r , and the radius of curvature $R(i)$:

$$N = r/d \quad (4)$$

$$\theta_{1} = \arcsin\left(\frac{r \cdot \sqrt{\frac{i-1}{N}}}{R(i)}\right) \quad (5)$$

$$\theta_{2} = \arcsin\left(\frac{r \cdot \sqrt{\frac{i}{N}}}{R(i)}\right) \quad (6)$$

In summary, the derivation of the three determinants of the Fresnel configuration highlights the lens radius (r), focal length (f), and ring spacing (d) as essential parameters. Notably, the lens refractive index (n) is solely dependent on the material properties used for lens fabrication, remaining unaffected during the design process. Therefore, the lens radius (r), focal length (f), and ring spacing (d) are critical design parameters for the configuration of the Fresnel lens.

4.2. Solar spectrum modeling

The Earth's atmospheric layer significantly affects the refraction, reflection, and absorption of sunlight as it reaches the surface, leading to the establishment of solar spectrum data standards collectively referred to as the air mass (AM) standard. These standards include various classifications, including AM0, AM1.0, and AM1.5 [49,50], which differ based on environmental conditions and usage requirements. Among these, the AM1.5 standard is the most widely utilized. In terms of standardization, AM1.5 has become the normative benchmark of the International Electrotechnical Commission (IEC), the U.S. National Renewable Energy Laboratory (NREL), and other authoritative organizations. In experimental contexts, AM1.5 is less challenging to replicate in laboratory conditions, as most commercial experimental equipment can deliver light output consistent with the AM1.5 standard, facilitating validation of scientific theories at the experimental level. Thus, this study adopts the AM1.5 data standard for solar spectrum simulation, referencing the national standard GB/T 17683.1-1999, drafted by the China Institute of Standardization and Information Classification and Coding (CISC), for specific data.

The solar spectrum includes a wide array of wavelengths, and a comprehensive modeling of all wavelengths is overly complex and generally unnecessary. Therefore, a computational simplification of the solar spectral data is essential to create a usable data set for modeling. In the spectral data standard, the wavelength values of individual sub-light

rays range from 0.305 μm to 0.990 μm , with neighboring light rays typically exhibiting wavelength differences of less than 0.03 μm , while larger differences occur only in specific wavelength regions. Thus, this research employs a computational simplification of the spectral data utilizing segmented integration; specifically, the median value of each five wavelength segments is designated as the representative wavelength for that segment, with the segmented irradiation intensity integral value serving as the representative irradiation intensity, and the segmented weight integral value as the weight representative for the light in that segment. The resulting solar spectrum data table is presented in Table 2.

4.3. Simulation environment

To investigate the simulation and optimization of the parameters of the polythermal module, this research utilizes TracePro software to model key process elements, including the light source, lens, and mold. Initially, sunlight is treated as parallel rays; thus, a lattice point light source model is employed for the light source in this simulation. Furthermore, given the circular shape and limited size of the lens used in this experimental system, the size of the light source distribution is set to be slightly larger than the lens's circular area, specifically with a radius of 600 mm. The energy of the light source distribution surface is uniform, meaning that a larger acceptance surface of the lens allows for the reception of a greater number of light rays, thereby increasing the energy collected. The standard value of solar irradiation intensity is established as 1361 W/m^2 , referencing the ASTM E490–22 standard. Lastly, the simulated wavelengths and weights for each type of light within the light source are configured based on the spectral simulation detailed in Table 2. In summary, the simulation research models 6504 light rays, with a total luminous flux emitted of 1539.25 W, as illustrated in Fig. 9.

Upon completion of the light source modeling, the physical components required for this simulation, primarily the lens and mold, must also be modeled. The lens, a critical component of this research, is characterized by three main parameters: radius, ring spacing, and focal length. As depicted in Fig. 10, the Fresnel lens module is directly invoked within TracePro for modeling purposes. The diameter and ring spacing of the lens can be directly defined through parameter designations. The focal length, however, is set indirectly via the object distance and image distance parameters. As a convex lens, the Fresnel lens adheres to the Gaussian lens formula:

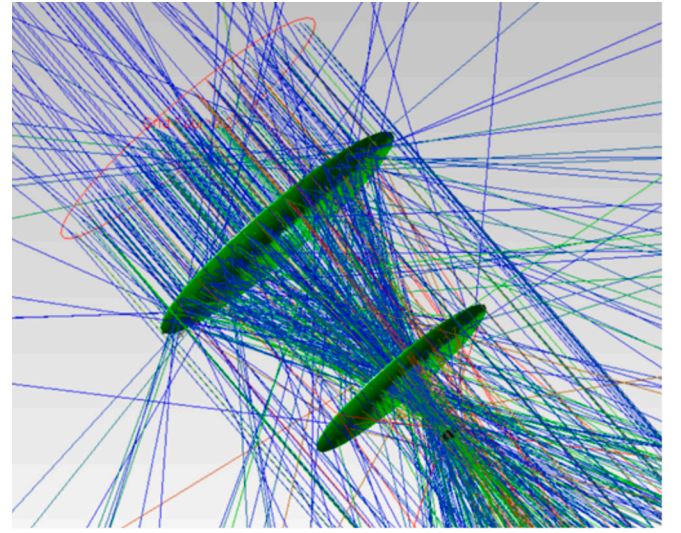


Fig. 9. Geometric optics tracking.

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \quad (7)$$

where f is the focal length of the lens and u and v are the image and object distances of the lens, respectively.

The molds utilized in this research are conventional cylindrical graphite molds featuring perforated holes at the base for holding the simulated lunar regolith and indenter. As illustrated in Fig. 10, the mold is constructed in TracePro with a tubular cylinder design. The wall thickness of the mold is set at 10 mm, and the radius of the hole at both the top and bottom is also established at 10 mm, ensuring continuity in size. Therefore, the diameter at the bottom of the mold is calculated as $2 \times 10 + 2 \times 10 = 40$ mm, with the mold's length subsequently set to 60 mm to finalize its dimensions and shape.

5. Performance analysis and discussion

Quantitative evaluation of concentration performance using irradiance, luminous flux, and energy efficiency metrics, with parametric studies on lens design variables.

5.1. Evaluation indicators

Irradiance refers to the amount of light power received per unit area, commonly measured in W/m^2 . In optical systems, particularly in imaging and illumination applications, understanding the focused optical irradiance, as well as its maximum, minimum, and average values, is essential for assessing system performance.

Total luminous flux is a vital metric for evaluating optical energy utilization, defined as the total energy value of light gained per unit of time, based on the intensity of light irradiation across various areas. This metric is typically measured in watts (W).

Energy utilization efficiency quantifies the proportion of energy in the light after incidence and ejection, reflecting energy losses within various physical components due to complex optical behaviors. In this research, two types of energy utilization rates are calculated:

- (1) Total energy utilization efficiency: This is the ratio of the luminous flux received by the mold to the total energy emitted by the light source, providing a global perspective on energy loss within the system.
- (2) Lens focusing energy utilization efficiency: This ratio compares the light flux received by the mold to the total energy emitted by

Table 2
Simulated solar spectral data.

Wavelength (μm)	Irradiance intensity integral(W/m^2)	Weights
0.38	46.17	33.68
0.43	106.02	59.85
0.48	184.88	78.86
0.53	262.38	77.5
0.61	408.52	146.14
0.71	502.21	93.69
0.7575	549.48	47.27
0.816	613.49	64.01
0.88	688.07	74.58
0.937	710.2	22.13
1.04	797.49	87.29
1.137	818.57	21.08
1.29	881.99	63.42
1.4625	890.12	8.13
1.558	914.19	24.07
1.646	946.48	32.29
1.92	955.48	9
2.065	967.68	12.2
2.36	987.19	19.51
2.973	989.85	2.66
3.204	991.66	1.81
3.573	998.79	7.13

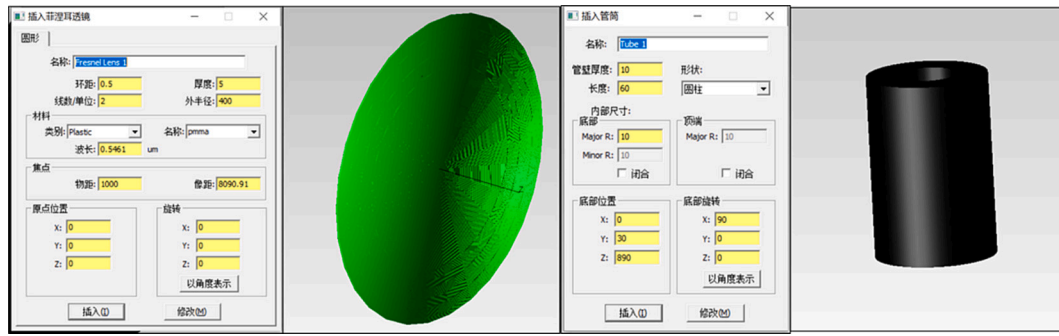


Fig. 10. Physical modeling.

the lens, focusing specifically on energy loss attributable to the lens's optical properties.

Optical energy wavelength distribution is determined by calculating the luminous flux and the number of light rays of different wavelengths. This distribution is particularly valuable for applications involving light with uneven energy distribution (e.g., sunlight). It allows for a detailed analysis of energy loss and residuals of various wavelengths of light as they pass through complex systems, facilitating further insights into the performance of precision optical systems such as lenses.

5.2. Effect of single design parameters on concentration performance

Parametric sensitivity analysis isolating ring spacing, focal length, and lens radius impacts on spot irradiance, energy utilization, and spectral distribution.

5.2.1. Effect of ring spacing

This research investigates the focusing performance of lenses with a radius of 400 mm and a focal length of 890 mm, specifically examining the differences in performance across various ring spacings ranging from 0.1 to 0.9 mm. The light irradiation intensity of the lens after focusing under these different ring spacing conditions is presented in Table 3. The data indicate that, despite variations in ring spacing, the final size of the focus remains consistent at approximately 30 mm. An analysis of the irradiation intensity reveals a minor effect of ring spacing. Notably, the maximum light irradiation value increases with larger ring spacing, with the optimal result occurring at 0.9 mm, yielding $2.760 \times 10^6 \text{ W/m}^2$. In contrast, the minimum light irradiation values for both 0.1 mm and 0.9 mm ring spacings demonstrate superior performance, significantly enhancing the other three groups by an order of magnitude, reaching the 10^{-13} level. However, when examining the average light irradiation values, changes in magnitude are minimal. Between 0.3 mm and 0.7 mm ring spacings, each 1 mm increase results in a gradual decline in average light irradiation intensity, while the two extreme values of 0.1 mm and 0.9 mm exhibit anomalous trends. Fig. 11 illustrates the spot light irradiation experienced by the mold receiving surface in the simulation environment. It should be noted that, due to the minimal effect of ring spacing on light irradiation intensity, visual differences among the five groups are subtle; thus, Fig. 11 displays only the case of 0.5 mm ring

spacing. The light spots exhibit an irregular shape on the mold's surface, characterized by a concentration of energy in the central region and a dispersion of energy at the edges. This observation arises from the combined effects of the lenses' optical behaviors for different wavelengths and the curvature of the mold surface.

In addition, Table 4 illustrates the numerical variations in luminous flux across different ring spacings. The modeling of the light source indicates that the total luminous flux emitted in the simulation environment of this research is fixed at 1539.25 W. Notably, larger lenses capture more luminous flux, with all lenses used in this section having a radius of 400 mm, thus receiving a consistent luminous flux of 833.06 W. Further analysis of each group reveals that the overall energy utilization for each group ranges from 26 % to 27 %. This limited energy utilization can be attributed to the restricted size of the lens receiving surface and energy loss during lens transmission. After accounting for the size of the lens receiving surface and focusing solely on the losses due to reflection, refraction, and diffraction, the total energy utilization for each group remains between 49 % and 50 %. In terms of variability analysis, the results diverge significantly from those observed in the prior light irradiation analysis. The group with a 0.1 mm ring spacing, which performed exceptionally well in light irradiation, ranks second to last in energy utilization. This suggests that the Fresnel lens with 0.1 mm ring spacing, despite its advantages in localized light irradiance, experiences greater overall energy loss. Conversely, the 0.3 mm ring spacing lens not only achieves optimal performance in the average light irradiance but also reaches maximum energy utilization efficiency. The 0.9 mm ring spacing follows closely, while the 0.7 mm ring spacing exhibits the lowest values for both average light irradiance and energy utilization efficiency, creating a substantial gap with the other four groups, with an efficiency of only 49.418 %.

To elucidate the effect of lens parameter changes on the wavelength distribution of light, this study also conducted energy statistics on the wavelengths received by the mold. Table 5 presents the wavelengths exhibiting significant energy differences. The influence of varying ring spacings on the wavelength distribution is predominantly concentrated in the short-wave band, particularly below $1.92 \mu\text{m}$. A detailed numerical analysis reveals that the 0.1 mm ring spacing demonstrates the most balanced performance, positioning itself in a median range among all wavelengths. In contrast, the 0.7 mm and 0.9 mm ring spacings underperform; the 0.7 mm spacing records the lowest values in the 0.38 and $1.559 \mu\text{m}$ wavelength bands, while the 0.9 mm spacing also achieves the lowest values in the 0.38, 0.816, and $1.4625 \mu\text{m}$ wavelength bands. However, it does reach maximum values in two other wavelength bands. The 0.3 mm ring spacing continues to demonstrate superior performance, achieving maximum values in two wavelength bands and a minimum in one.

5.2.2. Effect of focal length

This research investigates the differences in light focusing performance of a frontal lens with a radius of 550 mm at focal lengths ranging from 900 mm to 1300 mm. The light irradiation intensity following lens

Table 3
Spot irradiance data at different ring spacings.

R(mm)	d(mm)	Spot irradiance(W/m ²)		
		Max	Min	Mean
400	0.1	2.619×10^6	2.601×10^{-13}	7.621×10^4
	0.3	2.573×10^6	2.631×10^{-14}	7.623×10^4
	0.5	2.583×10^6	2.347×10^{-14}	7.618×10^4
	0.7	2.703×10^6	5.529×10^{-14}	7.617×10^4
	0.9	2.760×10^6	1.935×10^{-13}	7.621×10^4

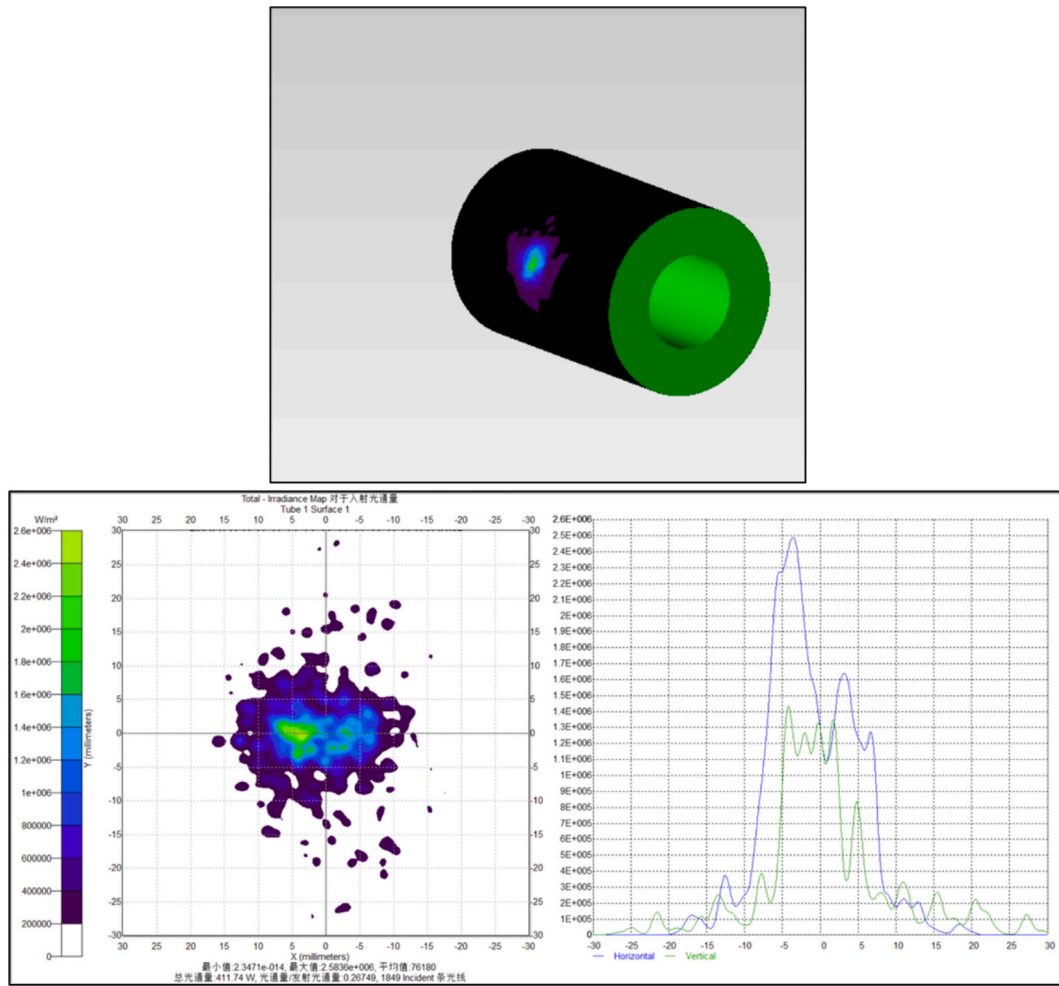


Fig. 11. $R = 400$ mm, $f = 890$ mm, $d = 0.5$ mm lens spot irradiance map.

Table 4

Luminous flux and energy efficiency at different ring spacings.

d (mm)	Emission (W)	Lens surface (W)	Mold surface (W)	Total energy utilization efficiency	Lens energy efficiency
0.1	1539.25	833.06	412.04	26.769 %	49.461 %
0.3		833.06	412.31	26.786 %	49.493 %
0.5		833.06	412.18	26.778 %	49.478 %
0.7		833.06	411.68	26.745 %	49.418 %
0.9		833.06	412.31	26.786 %	49.493 %

Table 5

Values of wavelength-dependent differences in luminous flux received by the mold at different ring spacings.

Wavelength (μm)	0.1 mm	0.3 mm	0.5 mm	0.7 mm	0.9 mm
0.38	6.385	6.405	6.366	6.327	6.327
0.816	26.203	26.203	26.203	26.203	25.871
1.137	9.187	9.187	9.187	9.187	9.297
1.4625	4.715	4.639	4.784	4.754	4.606
1.558	15.809	15.863	15.866	15.544	15.721
1.646	21.594	21.861	21.495	21.429	21.464
1.92	6.334	6.334	6.334	6.426	6.334

focusing at various focal lengths is presented in Table 6, while Fig. 12 illustrates the distribution of light irradiation in the simulation environment under the three focal lengths. From the perspective of spot size,

it is evident that an increase in focal length results in a gradual enlargement of the spot size, particularly in the longitudinal axis direction, with both sputtering range and breadth significantly enhanced. This increase is likely due to longer travel distances for all light rays to reach the mold, intensifying the effects of reflection, diffraction, and other undesirable phenomena. Numerically, Table 6 provides relevant quantitative data. Although the previous section indicated a gradual increase in spot size, this does not lead to excessive energy dispersion. As the focal length increases, both the average and maximum values of optical irradiance correspondingly enhance, though the enhancement of maximum irradiance is markedly uneven. The increase in maximum light irradiance from 1100 mm to 1300 mm is only one-fifteenth of the increase observed when transitioning from 900 mm to 1100 mm, indicating that while an increase in focal length can enhance maximum light irradiance, it is more constrained by other factors.

Table 7 further quantifies the differences in luminous flux and energy utilization efficiency for varying focal lengths. The data reveals a clear monotonic trend: as focal length increases, both the total energy

Table 6

Spot irradiance data at different focal lengths.

R(mm)	f(mm)	Spot irradiance(W/m^2)		
		Max	Max	Max
550	900	2.751×10^6	2.288×10^{-13}	8.835×10^4
	1100	3.305×10^6	9.830×10^{-14}	1.082×10^5
	1300	3.316×10^6	9.282×10^{-14}	1.319×10^5

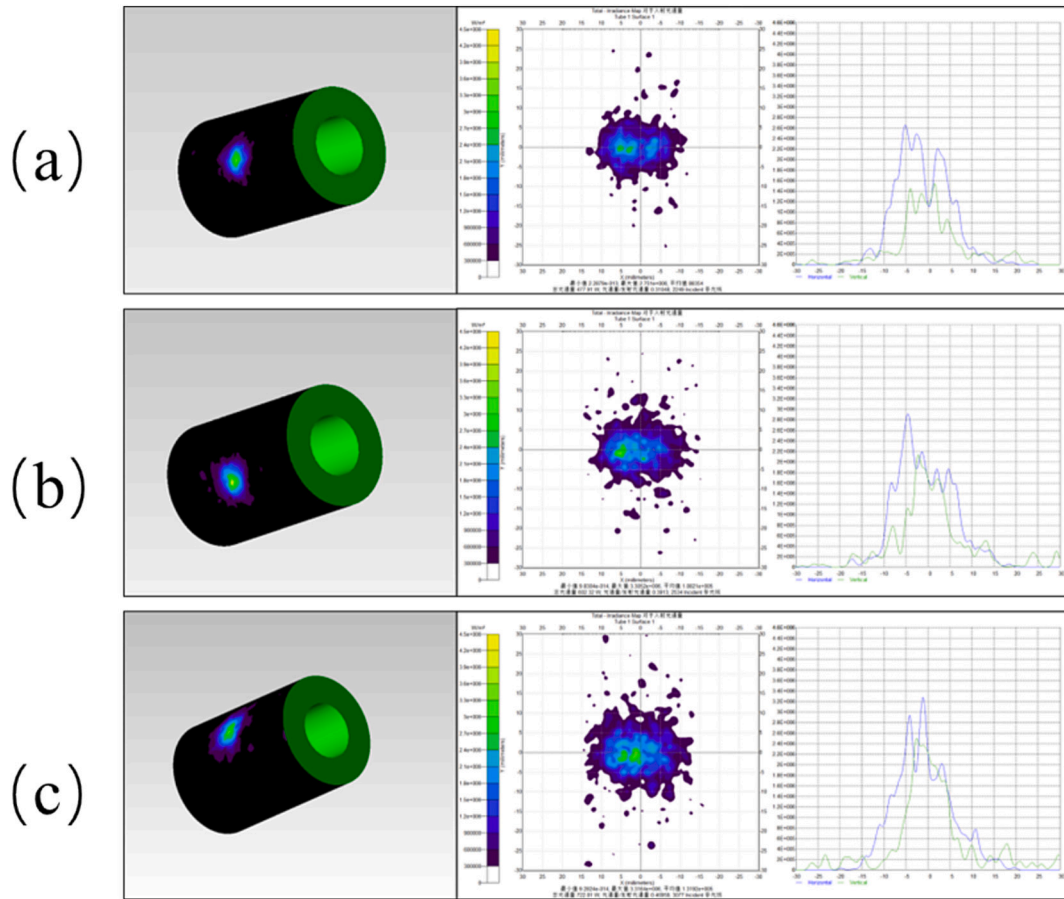


Fig. 12. Lens spot irradiance map at different focal lengths. (a)f = 900 mm; (a)f = 1100 mm; (a)f = 1300 mm.

utilization efficiency relative to the emitted luminous flux and the lens focusing energy utilization efficiency corresponding to the received luminous flux on the mold surface are simultaneously improved. Analyzing the magnitude of change, the total energy utilization efficiency uniformly rises, with each 200 mm increase in focal length yielding an approximate 8 % increase. The lens focusing energy utilization efficiency also improves, with each increase contributing approximately 9 % to 10 %.

Table 8 specifies the wavelengths at which significant energy differences occur. Unlike the effect of ring spacing, all 21 wavelength bands from 0.315 μm to 2.36 μm are notably affected by changes in focal length, with only a few ultra-high wavelength bands remaining unaffected. Numerical analysis reveals that, aside from the 0.315 μm wavelength band in the first row of Table 8, all other bands experience a significant increase in energy projected onto the mold surface as focal length increases. The middle wavelength bands exhibit particularly pronounced increases, with several wavelengths showing enhancements of approximately 10 W. Given that the middle wavelength band constitutes the largest and most concentrated region of the full spectrum, the combined effects lead to a substantial enhancement of energy on the illuminated surface of the mold, corroborating the findings from the

Table 8

Values of wavelength-dependent differences in luminous flux received by the mold at different focal lengths.

Wavelength(μm)	900	1100	1300
0.315	0.028	0.021	0.021
0.34	0.343	0.297	0.446
0.38	12.041	7.835	8.708
0.43	23.389	29.019	36.498
0.48	31.233	39.623	49.257
0.53	32.612	41.397	49.146
0.61	64.018	85.036	102.716
0.71	42.999	58.989	67.392
0.7575	22.259	29.276	35.988
0.816	29.954	40.324	48.410
0.88	35.591	47.773	54.855
0.937	10.693	14.178	16.858
1.04	45.366	55.029	68.346
1.137	11.054	13.736	18.169
1.29	35.775	43.690	56.066
1.4625	5.302	6.287	7.885
1.558	17.396	21.410	24.641
1.646	26.104	30.988	36.655
1.92	8.753	10.324	11.872
2.065	11.836	14.717	17.946
2.36	11.830	12.579	13.211

Table 7

Luminous flux and energy efficiency at different focal lengths.

f (mm)	Emission (W)	Lens surface (W)	Mold surface (W)	Total energy utilization efficiency	Lens energy efficiency
900		1307.30	478.83	31.108 %	36.627 %
1100	1539.25	1307.30	602.77	39.160 %	46.108 %
1300		1307.30	725.33	47.122 %	55.483 %

relevant prior data. Conversely, the 0.315 μm wavelength band represents a counter-trend, where increased focal length makes it increasingly difficult for light of this wavelength to reach the mold.

5.2.3. Effect of radius

In this study, the light concentration performance of Fresnel lenses

with radii of 250 mm, 400 mm, and 550 mm is analyzed to investigate the effect of varying radii on light concentration. Table 9 presents the light irradiation values of Fresnel lenses with different radii after concentration. The data indicate a consistent trend in both the maximum and average values of light irradiation: as the lens radius increases, the intensity of light irradiation at the focused spot also increases. The variations in these two indicators are relatively uniform; for the maximum value of light irradiation, every 150 mm increase in radius corresponds to an approximate rise of $0.8 \times 10^6 \text{ W/m}^2$. Similarly, for the average value, there is an increase of about $2.6 \times 10^6 \text{ W/m}^2$ to $2.8 \times 10^6 \text{ W/m}^2$ with each 150 mm increment. Notably, the minimum value is achieved with the smallest lens radius of 250 mm. This phenomenon aligns with the explanation provided in Section 5.2.2, where the lens size affects the optical range, leading to reduced light dispersion and a slightly higher minimum irradiation intensity.

Fig. 13 further illustrates the radius effect on light irradiance from a graphical perspective. At a radius of 250 mm, the distribution of light around the lens spot exhibits a regular pattern characteristic of an Airy spot phenomenon. According to Fraunhofer diffraction theory, when a parallel beam of light passes through a circular aperture (such as that of a lens or telescope), the light rays bend and interfere at the edges of the aperture, resulting in a unique diffraction pattern—namely, an Airy spot—on the focal plane. As the lens size increases, the prominence of the Airy spot diminishes. Furthermore, within the central region of the spot, the temperature correspondingly rises with increasing lens radius, indicating greater energy concentration.

Table 10 quantitatively details the differences in luminous flux and energy utilization efficiency across varying radii. Unlike the previous sections, the lens radius significantly impacts energy utilization efficiency. The data clearly indicate that with a radius of 250 mm, the light energy received is limited to 221.57 W due to the small lens surface area; however, after focusing, the retained energy is still 200.89 W, resulting in an impressive focusing energy utilization efficiency of 90.667 %. Despite this, the total energy utilization efficiency remains the lowest at only 13.051 %. The energy performance of the other two lens groups is comparatively moderate. Nonetheless, an overall trend emerges: as the lens radius increases, total energy utilization efficiency generally improves due to a larger surface area capturing more light energy. However, the increased lens surface area also heightens the potential for reflection, diffraction, and other phenomena, leading to greater energy loss during these optical interactions and thereby reducing the energy efficiency at the lens focus.

Table 11 presents the wavelengths of light exhibiting significant energy differences. The associated figure indicates that in the long-wavelength band above $3.0 \mu\text{m}$, variations in radius parameters exert negligible influence. Conversely, across all bands from 0.34 to $2.36 \mu\text{m}$, an increasing lens radius correlates with enhanced energy levels in the corresponding bands. This is particularly evident in the short- and medium-wavelength bands around $0.61 \mu\text{m}$, where the energy carried by the light nearly multiplies. The energy concentration near mid-wavelengths is the highest in solar spectral data, leading to a notable increase in total energy. However, the $0.315 \mu\text{m}$ wavelength band displays a decreasing luminous flux with increasing radius, stabilizing at a value of 0.021.

In summary, the analysis of the aforementioned indicators reveals that lenses with larger radii demonstrate superior performance. From a microscopic perspective, the large radius lenses achieve a significant

advantage in the mid-wavelength portion of the spectrum, where energy concentration is most pronounced. Therefore, the metrics of luminous flux and irradiance for larger radius lenses are considerably higher than those of smaller radii. However, the larger size also introduces limitations, as the increased volume through which light propagates results in greater energy loss, ultimately leading to reduced energy utilization efficiency as the lens radius increases.

5.3. Validation analysis of double-lens performance

In this study, simulation results under the dual-lens configuration were analyzed and compared with the single-lens case based on quantitative metrics. As shown in Fig. 14, the energy distribution was first evaluated at the point of incidence when the dual-lens structure was stacked. At the initial stage, the energy entering the large lens was uniformly distributed as pre-set lattice dots. After passing through the large lens and being focused onto the smaller lens, this lattice distribution transformed into a highly regular circular pattern with an approximate radius of 100 mm. Due to the reduced size of the second lens, the energy became more concentrated, resulting in an intensity increase of approximately 10-fold. Within this circular energy region, a central high-energy zone was observed, surrounded by a peripheral ring of enhanced brightness. This peripheral feature results from light dispersion and diffraction, where different wavelengths experience varying refractive indices within the same medium, causing them to bend at different angles and focus at separate positions.

Subsequently, this section compares the light irradiance between dual-lens and single-lens conditions, utilizing the illuminated side of the sensor as the analytical surface. As illustrated in Table 12, both the maximum and minimum values for the dual lens configuration exceed the respective results of either sub-lens under single action. However, while the average value exceeds that of the small lens, a minor difference remains compared to the large lens. The light irradiance distribution map further clarifies this comparison, as depicted in Fig. 15. It is evident that in the outer edge sputtering of the spot, the dual lens results closely resemble those of the large lens in a single action. In contrast, the central region displays shapes akin to both single lens configurations, characterized by irregularities. Nonetheless, in terms of energy distribution, the central region of the dual-lens spot exhibits more pronounced brightness, consistent with previous analyses of maximum optical irradiance.

Additionally, attention is drawn to the light irradiance distribution curve on the right side of Fig. 15. It is apparent that along the horizontal axis (blue), the energy distribution of the dual lenses generally exceeds the results of both single lenses and exhibits a more pronounced bimodal phenomenon, with the highest peak slightly offset to the left of the center. In the longitudinal direction (green), the peak of the curve for the single action of the large lens is higher, while the dual lens performance resembles a synthesis of the performance of the two single lenses, with the resulting value positioned centrally between those of the two individual lenses.

Table 13 presents the luminous flux values. Given that a two-lens system has two lens surfaces, it is essential to consider the incident energies for both surfaces. Thus, the lens focusing energy efficiency of the dual-lens system requires separate analysis for the larger and smaller lenses. Initially, from an overall perspective, the dual-lens combination captures luminous flux, with its performance influenced by lens size. However, a comparison reveals that the energy difference between the small and large lenses is minimal, remaining below 15 W. The total energy utilization efficiency for the dual-lens system is calculated at 38.211 %, which is significantly lower than the large lens's efficiency of 39.160 %, although it exceeds the small lens by nearly 12 percentage points. An analysis at the lens level indicates that, following the first lens, the energy utilization of the double lens is nearly double that of any single lens. Even after light passes through the second lens, the energy utilization of the dual lens reaches 57.254 %, maintaining a substantial

Table 9
Spot irradiance data at different focal radius.

R(mm)	Spot irradiance(W/m ²)		
	Max	Max	Max
250	1.758×10^6	1.323×10^{-13}	3.805×10^4
400	2.583×10^6	2.347×10^{-14}	7.618×10^4
550	3.305×10^6	9.830×10^{-14}	1.082×10^5

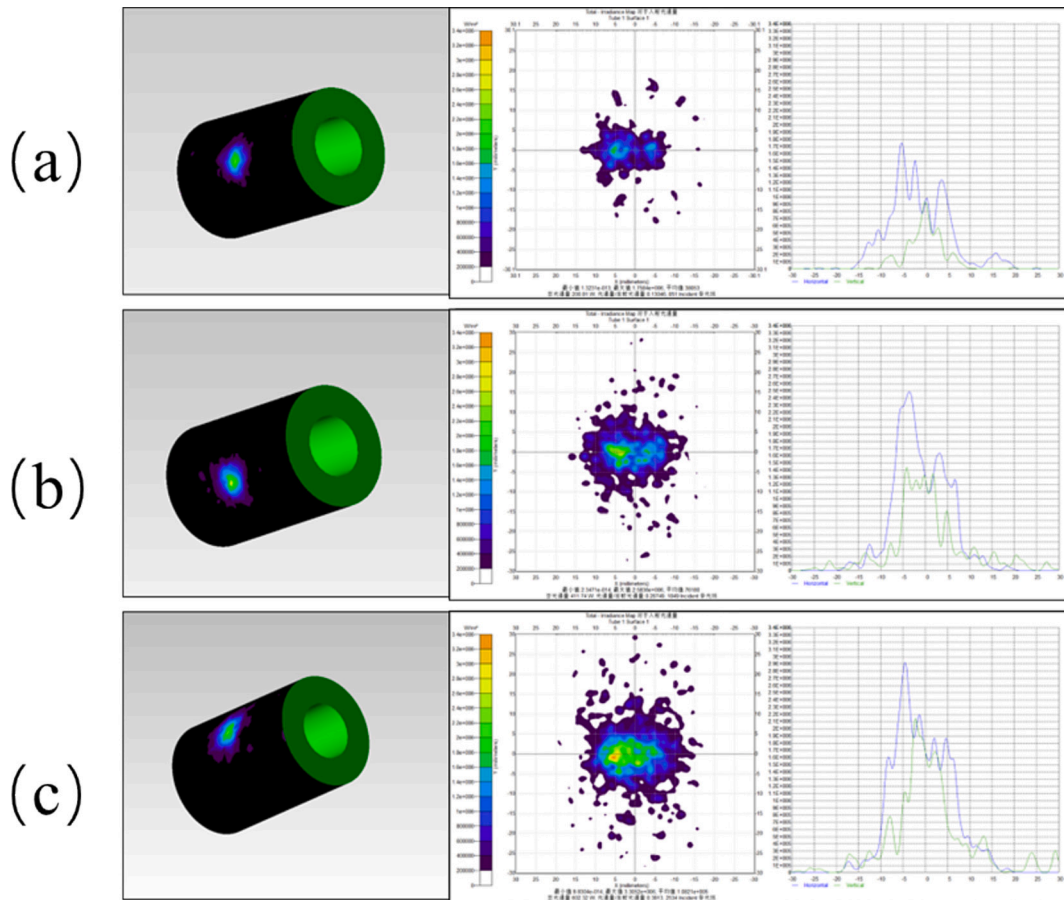


Fig. 13. Lens spot irradiance map at different radius. (a) $R = 250$ mm; (a) $f = 400$ mm; (a) $f = 550$ mm.

Table 10

Luminous flux and energy efficiency at different radius.

R (mm)	Emission (W)	Lens surface (W)	Mold surface (W)	Total energy utilization efficiency	Lens energy efficiency
250	1539.25	221.57	200.89	13.051 %	90.667 %
400		833.06	412.18	26.778 %	49.478 %
550		1307.30	602.77	39.160 %	46.108 %

advantage over any single lens. This suggests that the combination of two lenses, while slightly less efficient than the larger lens in total luminous flux, effectively reduces light energy loss due to lens focusing, allowing more energy to be directed onto the mold's surface.

To better understand the influence of the lens radius parameter on the wavelength distribution of light, energy statistics of the light received by the mold at sub-wavelengths were also conducted in this study. Table 14 illustrates the wavelengths exhibiting significant energy differences. In contrast to the single-lens simulation results, it is evident that all wavelengths of light are affected, to varying degrees, by the combined effects of the two lenses. Specifically, in the short-wavelength range of 0.315–0.61 μm , the double lens significantly enhances the energy ratio for this segment, showing an initial increase followed by a decrease in amplitude compared to the two single-lens configurations. In the mid-wavelength range of 0.71–2.065 μm , the advantages of the large lens become evident, although the dual lens lags by only a small numerical difference. Conversely, in the long-wavelength range, the dual lens again achieves superior performance.

In summary, the analysis of various indices indicates that the dual lens combination, due to its optimal matching effect, is more effective in

Table 11

Values of wavelength-dependent differences in luminous flux received by the mold at different focal lengths.

Wavelength(μm)	250	400	550
0.315	0.041	0.021	0.021
0.34	0.099	0.248	0.297
0.38	3.643	6.366	7.835
0.43	11.834	21.885	29.019
0.48	15.620	28.886	39.623
0.53	15.364	30.031	41.397
0.61	29.013	56.701	85.036
0.71	18.622	36.392	58.989
0.7575	9.399	19.345	29.276
0.816	12.730	26.203	40.324
0.88	14.836	29.763	47.773
0.937	4.403	9.292	14.178
1.04	17.374	37.116	55.029
1.137	4.198	9.187	13.736
1.29	12.648	29.656	43.690
1.4625	1.908	4.784	6.287
1.558	5.781	15.866	21.410
1.646	8.096	21.495	30.988
1.92	3.467	6.334	10.324
2.065	4.731	11.328	14.717
2.36	6.833	11.031	12.579

gathering energy from both short-wave and long-wave bands, while being slightly less effective than the large lens in the medium-wave band. This minor factor results in the dual lens combination being slightly weaker than the large lens in terms of luminous flux energy, with a negligible difference of only 15 W. However, regarding energy utilization efficiency, the dual lens combination significantly

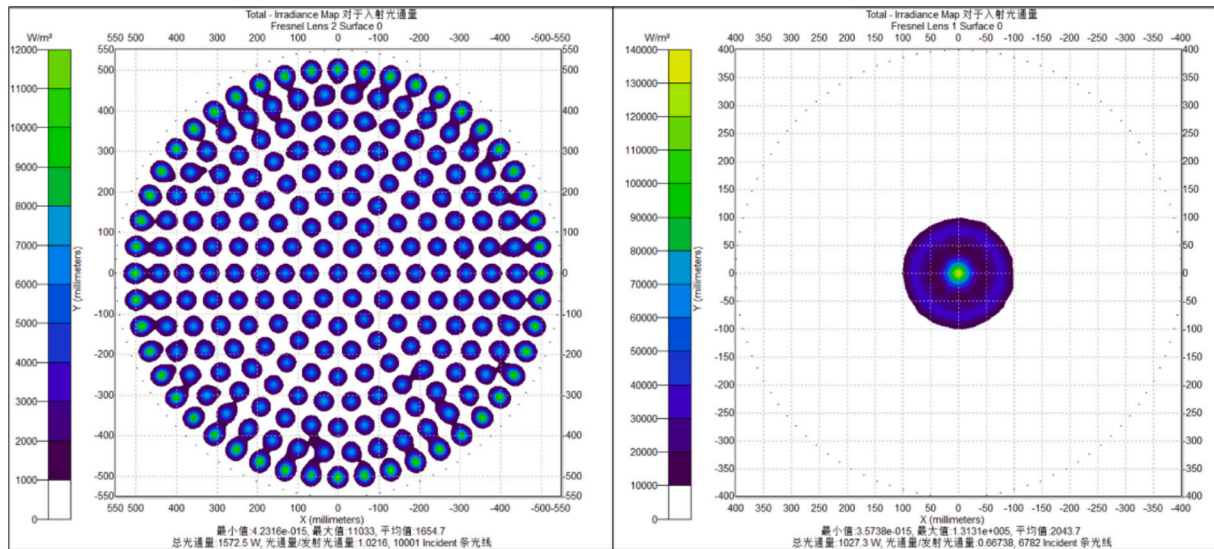


Fig. 14. Incident energy distribution for large and small lenses.

Table 12
Spot irradiance data.

Radius (mm)	Spot irradiance(W/m ²)		
	Max	Max	Max
400 + 550	3.506×10^6	3.081×10^{-13}	1.059×10^4
400	2.583×10^6	2.347×10^{-14}	7.618×10^4
550	3.305×10^6	9.830×10^{-14}	1.082×10^5

outperforms any single lens configuration, indicating reduced energy loss due to optical behavior. Moreover, the dual lens achieves optimal results concerning both maximum and minimum values of optical irradiation, suggesting its superior performance in focusing light energy.

6. Experimental tests and results analysis

Prototype deployment and validation workflow: machine assembly, electrical integration, HUST-1 simulant sintering tests, and performance characterization.

6.1. Machine integration and deployment

Following the machining of all components according to the design specifications, this research assembled the parts into a complete experimental machine. The assembly steps are illustrated in Fig. 16. The initial step is assembling the tilting motion mechanism, where the holes reserved for the lifting table are fitted with screws and four guide pillars, secured with a screw nut. A large pulley is attached externally to the screw nut; the rotation of this pulley drives the screw nut to lift the table for elevation adjustments. The large pulley connects to a smaller pulley via a belt drive, and the small pulley is linked to the output shaft of a stepping motor to provide driving force. The subsequent step is securing the vacuum tank to the bracket plate; the bottom of the vacuum tank features an array of eight screw holes, aligned with corresponding holes on the bracket plate to enhance the binding force. Next, the upper part of the mechanism is fixed to the bottom of the rotating platform and the cart. Similar to the vacuum tank assembly, the upper part of the rotating platform contains an array of screw holes, which are matched with corresponding holes in the turntable plate for secure attachment. The rotating platform is also secured at its four corners to the cart. The final assembly step is for the lens section. The lens is positioned within a quadrilateral frame formed by profile holders, followed by the

attachment of height adjustment modules on both sides, secured with fixing pegs. Finally, the profile frame containing the lens is affixed to the height modules.

Upon completing the mechanical assembly, it is necessary to design and lay out the electrical and control lines to ensure the system operates correctly. Fig. 17 depicts the actual configuration of the machine's electrical and control system. The motor reducer receives drive and direction signals, and the control lines for each drive converge at a hub connected to the control board. In addition, the control board features a limiter interface to prevent potentially hazardous movements that exceed safe limits. The power supply, located below the control board, converts conventional 220 V AC power into the 24 V DC required by the machine system, with a fan installed on the upper surface of the power supply for cooling. Ultimately, all wiring is consolidated within an electrical box, as illustrated in the figure.

The forming and monitoring components are centered on the deployment of the vacuum tank, which serves as the main structure. The primary content consists of various sensors and instrument panels, which are functionally divided into two categories: temperature measurement and vacuum monitoring, as shown in Fig. 18. The temperature measurement system utilizes the vacuum tank to house the graphite mold, leveraging graphite's excellent light and heat absorption properties to simulate lunar soil heating. The mold is positioned horizontally to ensure uniform heating of the internal cylindrical lunar soil. Asbestos insulation is placed between the mold and vacuum tank to maintain heat retention. A penetration hole near the simulated lunar regolith layer at the bottom surface of the graphite mold accommodates the thermocouple temperature measuring rod, ensuring accurate temperature readings. The thermocouple connects to a lead that interfaces with the internal contact point of the vacuum tank channel, while the external contact interfaces with a thermometer for temperature display. In contrast to the temperature measurement system, the vacuum measurement does not necessitate direct contact with the mold, only requiring interaction with any part of the vacuum tank channel.

6.2. Solar sintering experiment

Upon completion of machine deployment, it is essential to test the individual movements of the machine to verify adherence to design specifications. The debugging schematic for the rotary motion is depicted in Fig. 19, illustrating that the device can execute a full 360-degree circular motion around the center of the rotating platform, controllable via remote control or automatic control from the control

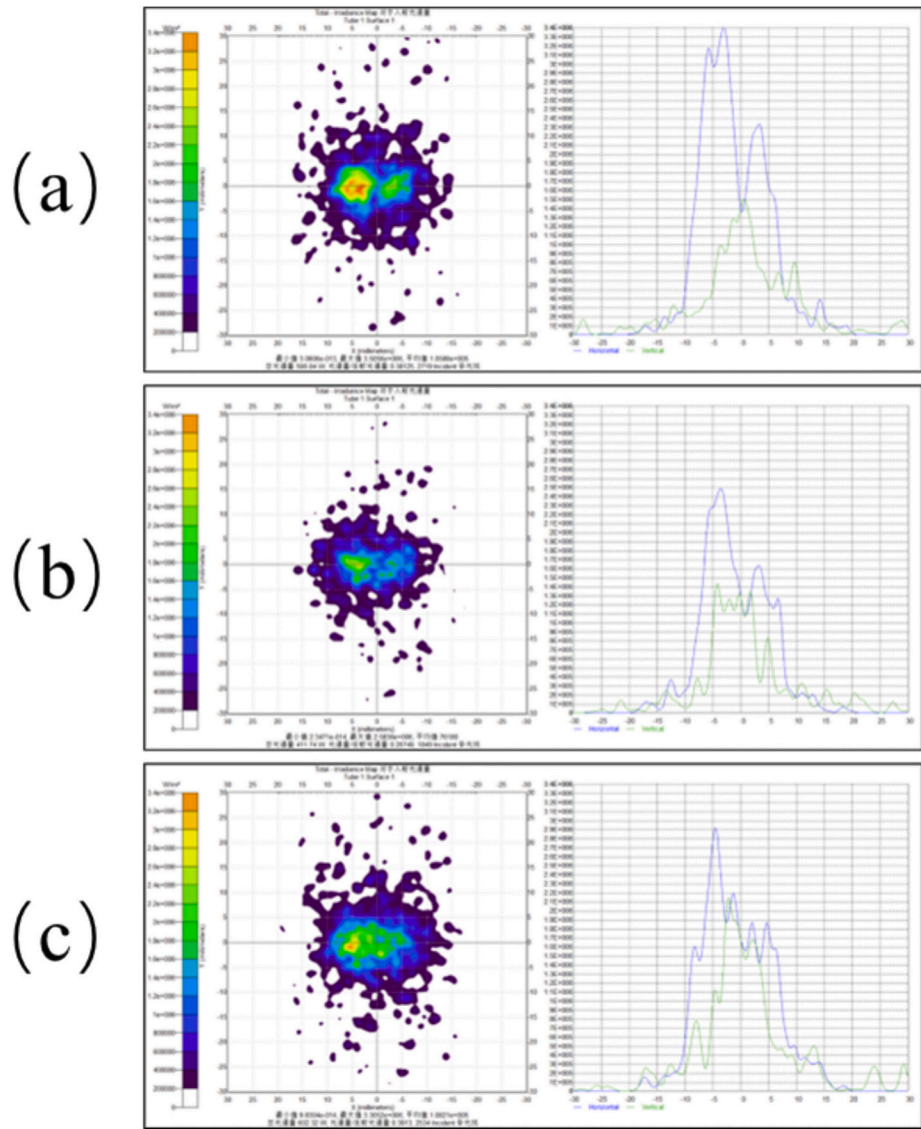


Fig. 15. Lens spot irradiance map at different focal lengths. (a)400 mm + 550 mm; (B)R = 400 mm; (C)R = 550 mm.

Table 13
Luminous flux and energy efficiency.

R(mm)	Emission (W)	Lens surface (W)	Mold surface (W)	Total energy utilization efficiency
400 + 550	1307.30/ 1027.30	588.17	38.211 %	73.582 %/57.254 %
400	833.06	412.18	26.778 %	49.478 %
550	1307.30	602.77	39.160 %	46.108 %

board.

Subsequently, this research conducts experiments utilizing the integrated and deployed experimental machine. Initially, the selection of experimental materials uses the HUST-1 model, a simulated lunar regolith. This model employs volcanic ash from Jilin Province as its raw material, which undergoes a specific process to remove biomass and is further treated through multiple stages, including grinding and sieving, to produce the HUST-1 simulated lunar regolith suitable for experimentation. Table 15 lists three types of lunar soil data. Among them, the Apollo 14 lunar soil data is the most widely used real lunar soil data in the field, and simulated lunar soil data from the United States (JCS), Japan (MKS and FJS), and China (CAS) are all based on it. The CAS

model of simulated lunar soil was developed by the Institute of Geochemistry of the Chinese Academy of Sciences and has been adopted by many scholars. The chemical composition of HUST-1 exhibits similarities to that of Apollo 14, with primary constituents such as SiO₂ and Al₂O₃. The composition of HUST-1 closely resembles the actual lunar soil retrieved by Apollo 14 and the CAS-1 simulated lunar soil widely used in scientific studies. Therefore, utilizing this model of simulated lunar soil as the experimental material can effectively reflect the characteristics of real lunar soil during the sintering process, possessing considerable theoretical and engineering significance.

In this section, a rigorous experimental procedure is employed for the solar mold sintering experiment of simulated lunar regolith, as depicted in Fig. 20. The initial step is powder molding, which requires pretreatment of the simulated lunar soil (See Fig. 21.), including sieving and drying, to ensure consistent particle size and the absence of moisture, sand, gravel, and other impurities. Following pretreatment, the simulated lunar regolith powder is loaded into the mold and secured with indenters on both sides for compaction. Given the weak fusion bonding between lunar regolith powders, it is essential to apply axial pressure using a press to enhance compactness after loading. Once the machine is prepared, the electrical switch is activated, and the device autonomously performs ray tracing based on sensor signals to adjust the

Table 14

Values of wavelength-dependent differences in luminous flux received by the mold.

Wavelength(μm)	400 + 550	400	550
0.315	0.025	0.021	0.021
0.34	0.443	0.248	0.297
0.38	9.025	6.366	7.835
0.43	28.195	21.885	29.019
0.48	37.654	28.886	39.623
0.53	43.125	30.031	41.397
0.61	85.811	56.701	85.036
0.71	51.795	36.392	58.989
0.7575	29.040	19.345	29.276
0.816	40.048	26.203	40.324
0.88	44.541	29.763	47.773
0.937	14.067	9.292	14.178
1.04	54.262	37.116	55.029
1.137	12.812	9.187	13.736
1.29	42.541	29.656	43.690
1.4625	5.753	4.784	6.287
1.558	18.431	15.866	21.410
1.646	27.510	21.495	30.988
1.92	8.869	6.334	10.324
2.065	14.067	11.328	14.717
2.36	19.707	11.031	12.579
2.973	0.142	0.147	0.147
3.204	0.089	0.020	0.020
3.573	0.219	0.080	0.080

machine's directional orientation. Subsequently, following the focusing process, sunlight is projected onto the mold's surface within the canister to create a high-heat spot that facilitates sintering of the simulated lunar regolith. Due to the power limitations of sunlight, the heating process throughout the experiment is relatively slow, lasting several hours. During this sintering period, the sun's position continually shifts, and the experimental machine communicates environmental information to

the control board based on signals from photosensitive sensors. The control board periodically readjusts the system's orientation to ensure the light spots consistently illuminate the mold's surface.

6.3. Analysis of experimental results

Validation of machine functionality (auto-tracking accuracy, heating curves) and sintered brick properties (morphology, compressive strength).

6.3.1. Performance of experimental machine

The effect of solar auto-tracking during experimentation is analyzed in accordance with the specific conditions of the study. In the actual operation of the experiment, the preset functions of the manual remote control are effectively utilized, enabling manual control of both pitching and rotary motions, as well as seamless switching between manual and automatic functions. The automatic light chasing primarily serves to fine-tune the attitude of the experimental machine during both the initial and subsequent phases of the experiment, ensuring that the light spot remains focused on the mold's surface. In practice, the automatic light chasing demonstrates considerable effectiveness, maintaining a frequency of 1–2 min for refreshing. During the refresh process, the system's attitude can be adjusted based on real-time light sensing information, achieving precision within 1 degree during adjustments. Various moments in the experimental light-gathering process are illustrated in Fig. 22, clearly depicting the continuous clockwise rotation and movement of the device's bracket plate, while the mold's position remains relatively unchanged. This observation confirms the utility of the experimental machine's light chasing performance. It is noteworthy that, due to the strong light reflection from the quartz glass atop the vacuum tank, capturing images of the light spots on the mold is challenging when the tank lid is closed; thus, it is necessary to remove the lid to visually observe the light spots during the demonstration of the light

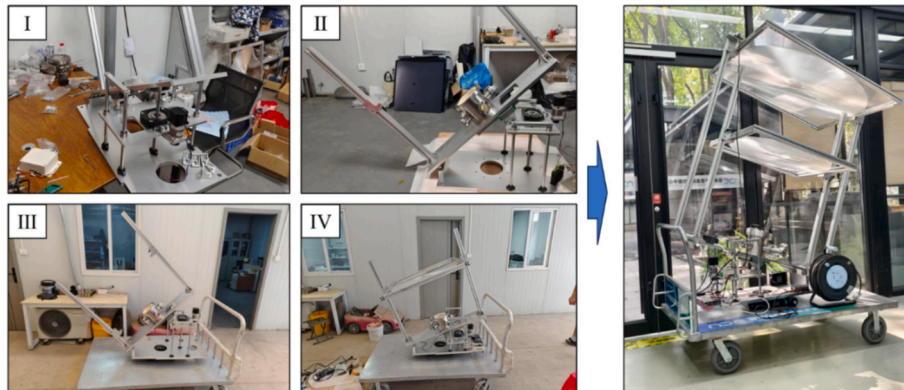


Fig. 16. Mechanical assembly schematic.

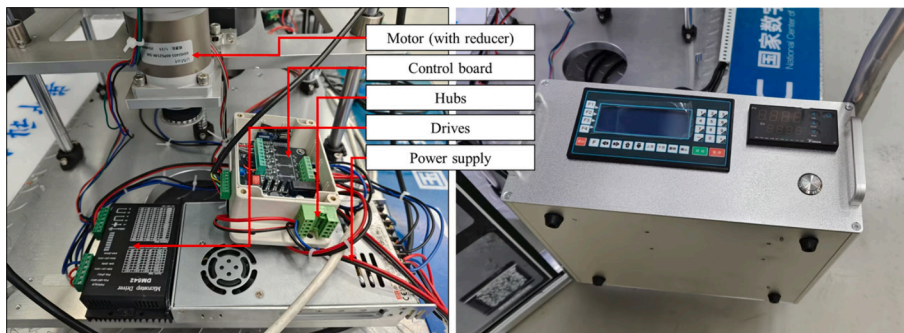


Fig. 17. Electrical and control layout.

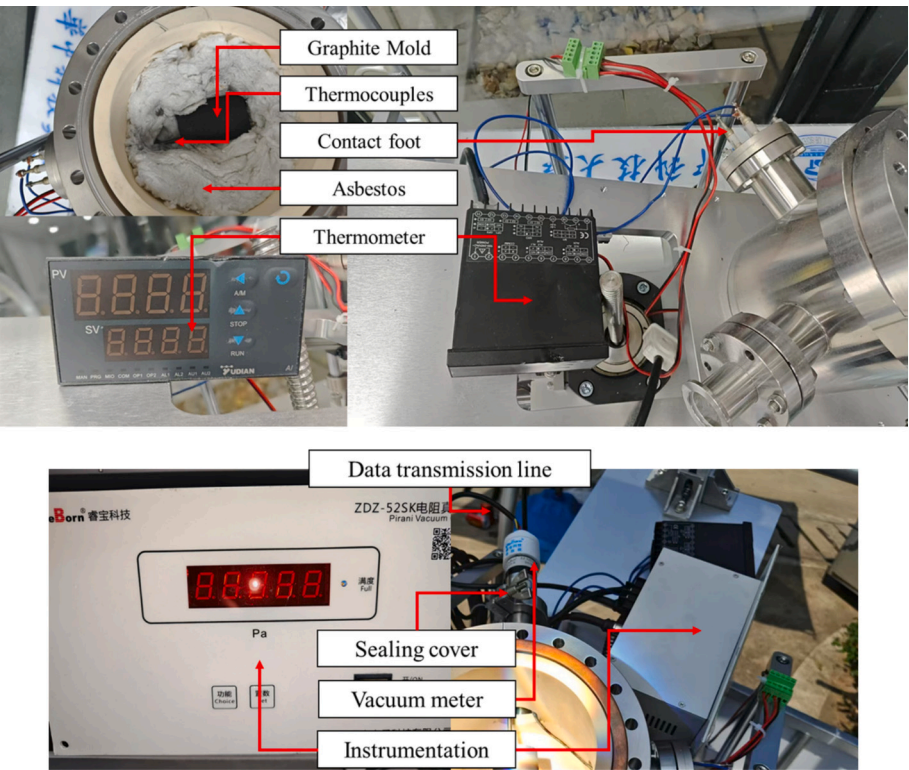


Fig. 18. Monitoring sensor installation.

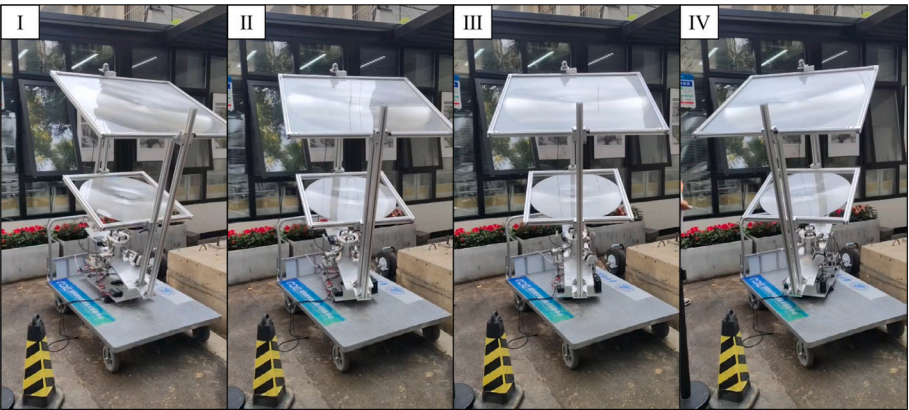


Fig. 19. Demonstration of the movement of the machine.

Table 15
Composition of Hust-1.

	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	CaO	MgO	Na ₂ O	K ₂ O	TiO ₂	P ₂ O ₅
Apollo 14	48.10	17.40	10.40	10.70	9.40	0.70	0.55	1.70	0.51
CAS-1	49.24	15.80	11.47	7.25	8.72	3.08	1.03	1.91	0.30
HUST-1	48.23	18.29	11.19	7.89	4.41	3.70	2.15	2.96	0.50

tracing function.

Subsequently, the heating performance of the entire experimental machine is analyzed according to the specific conditions of the experiment. During the experiment, a temperature point is recorded every 10 min, resulting in the temperature curve illustrated in Fig. 23. The temperature trend indicates that the mold experiences a rapid increase in temperature at the onset of light gathering, primarily due to the initial temperature of the mold and surrounding air being consistent at approximately 30 degrees Celsius. The initial energy is low, and the

sun's spot capacity is substantial, allowing the mold to absorb heat effectively, particularly during the first half-hour, where the temperature rises nearly 100 degrees every 10 min. As the experiment progresses, the continuous increase in mold temperature leads to a significant decrease in heat-absorbing capacity. After one hour, the mold temperature becomes increasingly difficult to raise, with the rate of warming dropping sharply to approximately 40 degrees per minute. When the mold temperature exceeds 600 degrees, further increases become challenging, and as it approaches 700 degrees, the temperature

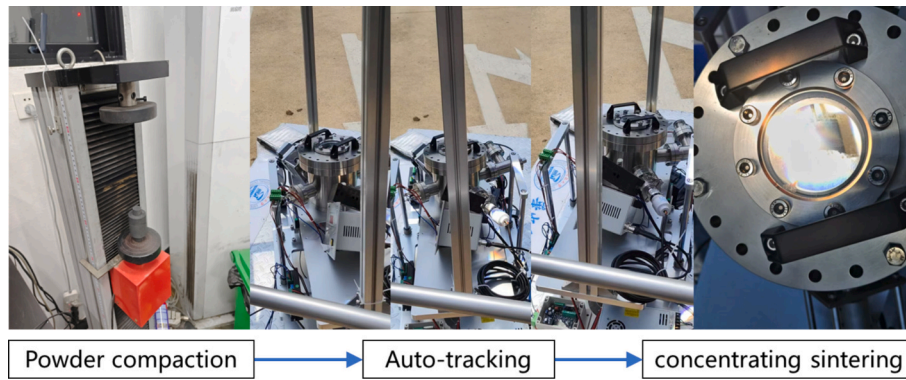


Fig. 20. Processes for concentrated light sintering.



Fig. 21. Particle morphology of HUST-1 [3].

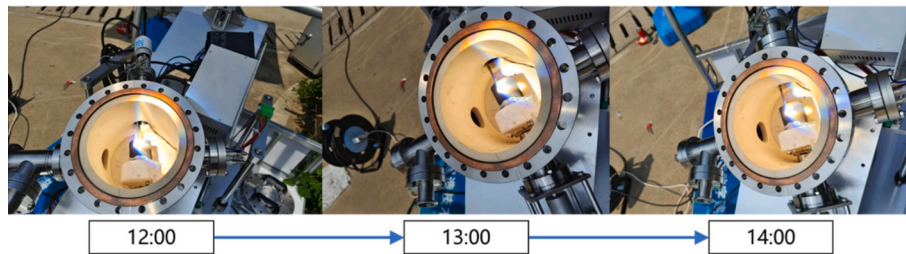


Fig. 22. Light tracing display.

rise per 10 min can be counted in single digits. Additionally, at temperatures above 700 degrees, external factors such as breezes and clouds contribute to fluctuations in mold temperature, resulting in a downward trend at certain times. The final temperature of the experiment ranges between 700 and 800 degrees, marking the experimental limit.

6.3.2. Performance of lunar bricks with regolith simulant

Following the sintering process in the experimental equipment developed in this research, the initial formation of lunar regolith components in the mold was achieved. Fig. 24 presents physical representations of the simulated lunar regolith components formed under both vacuum sintering and atmospheric conditions. Although the experimental temperature did not reach the ideal melting temperature for lunar regolith sintering, the simulated lunar regolith underwent high-temperature sintering, resulting in tightly bonded lunar regolith particles that formed a complete sample shape and exhibited a certain degree of mechanical strength. In terms of appearance, the vacuum-sintered samples are characterized by a black color, while the atmospherically-sintered samples display a slight gray hue, attributed to partial oxidation of some lunar regolith particles upon contact with oxygen, despite

the protective barriers of the molds and indenters. Furthermore, some powder may have escaped from the components due to incomplete sintering temperatures. Regarding dimensions, the formed lunar soil components are nearly cylindrical, with a diameter of approximately 20 mm at the base and a height of around 30 mm. However, specific dimensions vary among different components due to incidental errors during the powder loading and pressing processes. Subsequently, a compressive strength test of the sintered components was conducted in this research, revealing that the compressive strength of the vacuum-sintered components generally falls within the range of 0.4–0.8 MPa, while the strength of the atmospherically sintered components is slightly lower, approximately between 0.2 and 0.5 MPa.

6.4. Analysis of factors affecting sintering

Critical factors governing brick quality: atmospheric conditions (vacuum vs. air) and temperature gradients (500–800 °C), validated via SEM morphology.

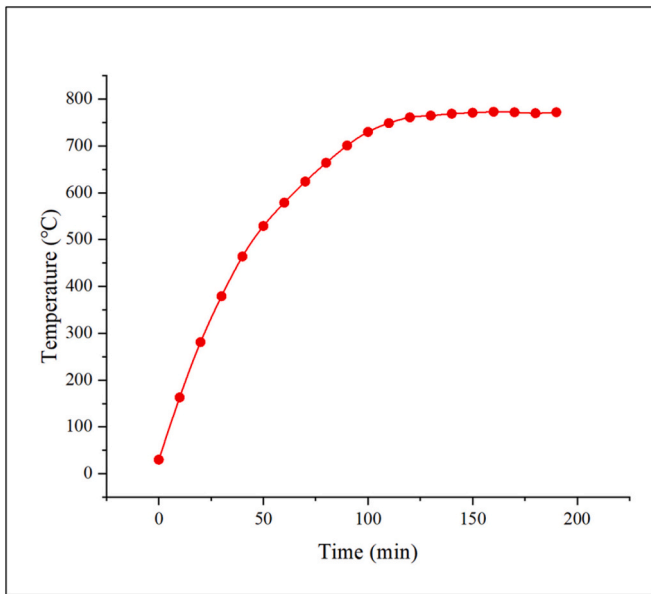


Fig. 23. Mold heating curve.

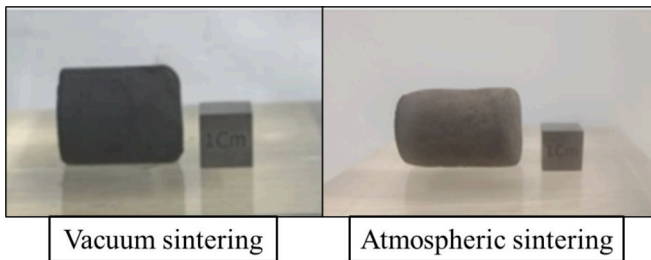


Fig. 24. Sintered components.

6.4.1. Effect of air conditions on sintering

In order to illustrate the effect of the properties related to the experimental system on sintering in this research, this section focuses on the air conditions, a critical performance factor, and examines the differences between vacuum and atmospheric-fired structural components. As depicted in Fig. 25, the particle micromorphology maps of HUST-1 powder, vacuum-fired structural components, and atmospheric-fired structural components obtained through scanning electron microscopy (SEM) are presented. Initially, the SEM images at the 20 μm scale were analyzed, revealing that the micro-particle morphology of both vacuum sintering and atmospheric sintering exhibited notable changes, with significant improvements in inter-particle densities. Upon comparing vacuum sintering and atmospheric sintering, the microscopic morphology and densities of the vacuum-sintered samples are markedly

superior to those of the atmospheric-sintered samples. Specifically, atmospheric-sintered samples display inferior sintering uniformity due to the interference of air during the sintering process, resulting in pronounced regional variability and the presence of large pores and cracks among various sintered regions. Conversely, while vacuum-sintered samples also exhibit irregular pores and particle bumps on their surfaces—attributable to residual air in the mold—the density and uniformity of the surface sintering are significantly enhanced due to the absence of substantial air interference, contributing to the higher strength of the vacuum-sintered components.

In summary, the vacuum environment integrated into the sintering system developed in this research can significantly enhance the densities and uniformities of the components during the sintering process. However, due to residual air remaining within the molds, certain pores and granular bumps persist in the microscopic morphology of the samples after vacuum sintering. Future improvements in the vacuum system and mold-making processes may further enhance the performance of the sintered structural components.

6.4.2. Effect of temperature on sintering

This section explores the effect of sintering temperature on the performance of the sintered parts. Fig. 26 presents the SEM images of the sintered structures under varying temperature conditions. The temperatures indicated in the figure represent a range rather than exact values, as fluctuations occur during the sintering experiments due to environmental factors such as light and airflow. The particle morphology illustrated in the figure demonstrates that, despite variations in sintering temperatures, all samples exhibited a clear trend toward densification compared to the original lunar regolith powder. At lower temperatures, specifically 500–600 $^{\circ}\text{C}$, sintering on the surface of the components is suboptimal, with densification occurring only in some regions, while a significant portion of the particles remains in a discrete state. The brightness and darkness variations in the SEM images indicate that the sintering uniformity of the components at 500–600 $^{\circ}\text{C}$ is also inadequate, characterized by numerous pitted areas. As the temperature increases to 700–800 $^{\circ}\text{C}$, the previously described conditions improve significantly; the majority of particles crystallize and fuse, resulting in a marked enhancement in the degree of densification. Additionally, sintering uniformity improves notably, with most regions of the simulated lunar regolith, excluding peripheral and localized areas, receiving superior sintering treatment, thereby enhancing the flatness of the components.

In conclusion, the experimental system developed in this research demonstrates a notable densification effect on the simulated lunar regolith, with higher sintering temperatures leading to significant improvements in both the degree of densification and the uniformity of the samples, as evidenced by the experimental results. Future iterations of the system can focus on enhancing heating and heat preservation capabilities to produce components with greater strength.

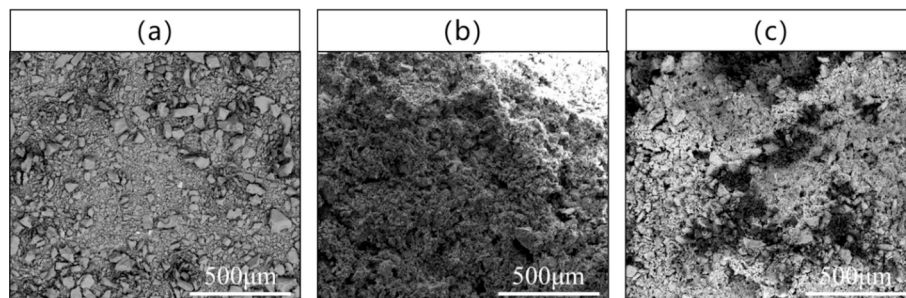


Fig. 25. Scanning electron micrographs of burnt structural parts under different air conditions: (a) HUST-1 powder; (b) vacuum burnt structural parts; (c) atmospheric burnt structural parts.

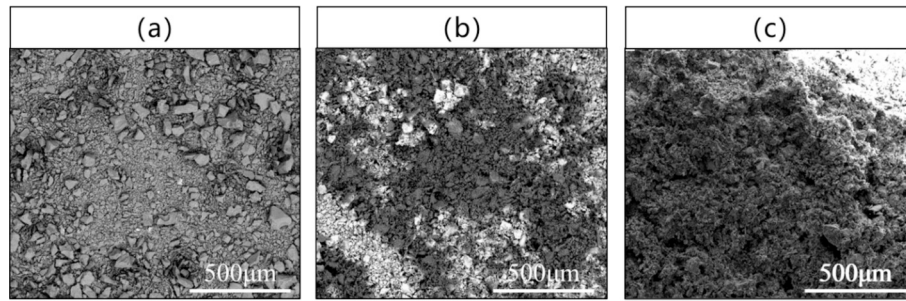


Fig. 26. Scanning electron micrographs of sintered structural parts at different temperature conditions: (a) HUST-1 powder; (b) sintering temperature: 500–600 °C; (c) sintering temperature: 700–800 °C.

6.5. Gap analysis

System limitations identification: Earth-Lunar environmental discrepancies, mold diversity needs, and scalability challenges for extra-terrestrial production lines.

6.5.1. Differences between the system proposed and previous research

Previously, researchers have explored Earth-based experimental systems for sintering lunar bricks. Compared with previous studies, the ground-based experimental system developed in this research has the following differences and improvements:

- (1) The technical approach used in this study is not powder bed fusion, but mold forming. The weak interlayer bonding force between lunar soil particles means that powder bed fusion either results in extremely thin components or requires the addition of extra material for reinforcement. The mold forming approach used in this research not only greatly reduces the difficulty of forming, but also allows for the formation of components of different shapes by changing the mold.
- (2) The Earth-based experimental system developed in this research has a high degree of automation. Previous research relied almost entirely on manual operation, whereas the experimental system proposed in this research can achieve automated sintering through the functions of the electrical system and tracking system, offering greater application prospects.

6.5.2. Limitations of earth-based systems

Although the work carried out in this paper has achieved certain sintering effects, there are still certain limitations in the research.

- (1) Simplified consideration of environmental factors. For typical environmental factors on the moon, such as low gravity, vacuum, and dust, this paper selected the vacuum factor, which has the greatest impact, as the key requirement for the design of this study. However, other environmental factors were not sufficiently considered, but their influence cannot be ruled out. For example, low gravity affects the movement of equipment and the binding of lunar soil particles, while dust poses a threat to the reliability of equipment. Future research could further consider these environmental factors to achieve a more robust experimental system.
- (2) Poor consistency between simulation and experimental metrics. The simulation work in this research involved multiple metrics, including luminous flux and light irradiance intensity. However, during the experimental process, it was difficult to measure the relevant metrics on the focused mold surface because the mold was sealed in a vacuum tank. Therefore, in the experimental stage, we mainly explored changes in easily measurable macro metrics such as temperature and component strength. Future

work could further refine this aspect to achieve unified analysis between the simulation and experimental stages.

- (3) Accuracy and uncertainty of the light-tracking system. The light-tracking system designed in this study mainly relies on simple circuit feedback control using photoresistors, which results in low accuracy and a certain degree of uncertainty. In the future, more accurate active control algorithms can be developed in combination with astronomical models to achieve better light-tracking effects.

6.5.3. Analysis of the gap between the earth system and the lunar system

The lunar brick sintering experimental system developed in this research is a preliminary technical exploration and equipment experiment, mainly developed under earth conditions. However, in the future, it will be inevitable to use automated equipment for construction activities on the moon. Therefore, this section analyzes the experimental system proposed in this research based on the differences between earth and lunar conditions:

- (1) The overall design of the experimental system is relatively unsophisticated, with insufficient focus on lightweight construction and miniaturization. The system design and prototype deployment emphasize functionality, resulting in components that are predominantly larger and heavier. In actual spaceflight conditions, the size and weight of the load can significantly increase capacity costs while introducing additional risks. Future work should prioritize precise design of internal components or the use of advanced aerospace-grade materials to achieve lightweight and miniaturized systems that better meet the demands of spaceflight.
- (2) Diversified mold design. In real construction scenarios, various component shapes are required. Therefore, to meet construction needs, molds of a single shape are insufficient. It is necessary to design diversified molds or convenient reconfigurable molds, and explore the differences in their sintering processes from an experimental perspective, thereby better meeting the actual needs of lunar construction.
- (3) To achieve complete sintering of lunar soil bricks, it is necessary to establish a complete production line. It is difficult to carry out large-scale lunar brick manufacturing with a single piece of equipment. In addition to further improving the experimental system proposed in this study, it is also necessary to combine it with supporting systems that can work together, such as a lunar soil powder delivery system and a lunar soil brick picking system.

However, there are still many favorable conditions for sintering under lunar conditions. For example, the angle between the lunar equator and the ecliptic is only 1.5 degrees, and one lunar day is approximately 30 Earth days. Therefore, the magnitude and rate of change of the solar altitude angle under lunar conditions are smaller than those on Earth, which also makes the equipment attitude more

stable during the sintering process. Additionally, the solar radiation intensity on the lunar surface is approximately 200 times that on Earth's surface, providing significantly greater energy input than on Earth, which facilitates heating during experiments.

7. Conclusions

Based on the analysis of environmental conditions and functional requirements, an automated ground-based experimental system, designated the USSS, was designed and developed for the solar sintering and shaping of simulated lunar regolith. This system is capable of automatically adjusting its orientation to align with solar motion and sintering the simulated lunar regolith within the mold by concentrating light and generating heat using Fresnel lenses. In addition, this research simulates and analyzes the system's heat-gathering module and critically evaluates its performance in heat-gathering sintering. Ultimately, this research integrates and deploys the experimental machine of the ground-based experimental system and conducts several solar concentrating sintering experiments to form simulated lunar regolith components with defined shapes and strengths, while also exploring avenues for optimization of the experimental machine's design.

However, the experimental system proposed in this paper is only a "first-generation" product, the purpose of which is to establish preliminary experimental capabilities for subsequent iteration and upgrading work. In the future, the structural design and thermal insulation performance of the equipment can be optimized based on experimental feedback, thereby achieving better heating capabilities and forming stronger lunar brick components.

CRediT authorship contribution statement

Cheng Zhou: Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yanan Lu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yuyue Gao:** Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yan Zhou:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Shifeng Wen:** Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Lieyun Ding:** Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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Data availability

Data will be made available on request.

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